

Climate Change and Mitigation Strategies

Introduction

Climate change refers to long-term changes in temperature and weather patterns, primarily driven by human activities such as burning fossil fuels, deforestation, and industrial emissions that increase greenhouse gases in the atmosphere. These changes lead to global warming, rising sea levels, extreme weather events, and disruptions to ecosystems and livelihoods. Addressing climate change is crucial to protect the environment, safeguard human health, ensure food and water security, and promote social equity. It also aligns with Sustainable Development Goal 13: Climate Action, which calls for urgent global efforts to reduce emissions, strengthen climate resilience, and integrate climate measures into policies and education to build a more sustainable future.

Learning Objectives

1. Analyze the causes and effects of climate change.
2. Examine global warming's impact and mitigation and adaptation strategies concerning SDG 13.

I. Causes of Climate Change

A. Natural Causes

1. Volcanic Eruptions

Large volcanic eruptions can release significant amounts of ash and gases like sulfur dioxide into the atmosphere. These particles can reflect sunlight away from Earth, leading to temporary cooling. However, over more extended periods, volcanoes release carbon dioxide (CO₂), a greenhouse gas contributing to warming.

2. Solar Radiation Variations

Changes in the sun's activity can vary the amount of energy the Earth receives from the sun. These natural solar cycles can cause short-term warming or cooling trends, but do not fully explain the rapid climate change observed in recent decades.

3. Natural Greenhouse Effect

The Earth naturally maintains its temperature through the greenhouse effect, where gases like water vapor, CO₂, and methane trap heat in the atmosphere. This process is essential for life, but when the concentration of these gases increases abnormally, especially from human activities, the greenhouse effect intensifies, causing global warming.

B. Human-Induced Causes

1. Burning of Fossil Fuels

Coal, oil, and natural gas combustion for electricity, transportation, and industry releases large amounts of CO₂ into the atmosphere. This is the primary driver of the enhanced greenhouse effect and global warming.

2. Deforestation

Trees naturally absorb CO₂ from the atmosphere. When forests are cleared for agriculture, logging, or urban development, this carbon absorption capacity is lost, and the carbon stored in trees is also released back into the atmosphere, increasing greenhouse gas levels.

3. Industrial Processes

Manufacturing and industrial activities emit various greenhouse gases, including CO₂ and fluorinated gases, during the production of cement, steel, chemicals, and other goods. These processes significantly contribute to atmospheric warming.

4. Agriculture and Livestock

Agricultural activities, particularly livestock farming, release methane (CH₄) through enteric fermentation (digestion in animals like cows) and from manure. Additionally, the use of nitrogen-based fertilizers releases nitrous oxide (N₂O), another potent greenhouse gas. These emissions play a major role in accelerating climate change.

II. Effects of Climate Change

A. Environmental Impacts

- Rising global temperatures

Global average temperatures have increased due to the enhanced greenhouse effect, leading to altered weather patterns, heatwaves, and shifting climate zones. These temperature changes disrupt ecosystems and biodiversity.
- Melting glaciers and polar ice caps

Warming temperatures accelerate the melting of glaciers and ice sheets in Greenland and Antarctica. This contributes significantly to sea level rise and reduces habitats for cold-adapted species such as polar bears and penguins.
- Sea level rise

As ice melts and ocean water expands due to warming, sea levels continue to rise. Low-lying coastal regions and small island nations face increased flooding, saltwater intrusion into freshwater sources, and loss of land.
- Increased frequency of extreme weather events

Climate change intensifies extreme weather events such as hurricanes, droughts, floods, and wildfires. These events have become more frequent and severe, causing widespread environmental and economic damage.

B. Social and Economic Impacts

- Threats to food and water security

Changes in temperature and precipitation disrupt agricultural productivity, reduce crop yields, and affect freshwater availability. This threatens global food supply and increases the risk of famine in vulnerable regions.
- Displacement and climate refugees

Sea level rise, extreme weather, and environmental degradation force communities to migrate. Climate-induced displacement is increasing, especially in coastal, arid, and disaster-prone areas.
- Health risks

Climate change worsens health problems by increasing heat-related illnesses, expanding the range of vector-borne diseases (e.g., dengue, malaria), and worsening air quality. Vulnerable populations (children, elderly, poor communities) are most at risk.
- Economic losses in agriculture, fisheries, and tourism

Extreme weather, habitat loss, and changing environmental conditions negatively affect major economic sectors. Crop and fishery declines reduce income and food supply, while tourism suffers from coral bleaching, extreme heat, and disaster impacts.

III. Global Mitigation and Adaptation Strategies

A. Mitigation Strategies

- Renewable energy (solar, wind, hydro)

Transitioning from fossil fuels to renewable energy sources reduces greenhouse gas emissions and decreases reliance on carbon-intensive energy systems. Solar, wind, and hydropower significantly lower CO₂ output and contribute to sustainable energy security.
- Energy efficiency and conservation

Improving energy efficiency in transportation, buildings, and industry helps reduce overall energy demand. Conservation practices—such as optimized appliance

use and efficient urban design—also lessen carbon footprints and reduce emissions across sectors.

- **Reforestation and afforestation**
Planting new forests (afforestation) and restoring degraded ones (reforestation) helps absorb atmospheric CO₂, enhance biodiversity, and stabilize ecosystems. Forests serve as major carbon sinks and contribute to climate regulation.
- **Carbon pricing and emissions trading**
Carbon pricing mechanisms, including carbon taxes and cap-and-trade systems, encourage industries to reduce emissions by assigning a financial cost to carbon pollution. These policies incentivize cleaner technologies and more sustainable business practices.

B. Adaptation Strategies

- **Climate-resilient infrastructure**
Infrastructure designed to withstand climate impacts—such as flood-resistant roads, elevated buildings, and resilient power grids—reduces vulnerability to disasters and supports long-term development.
- **Early warning systems**
Advanced monitoring technologies and communication networks help communities anticipate extreme weather events. Early warnings save lives, reduce economic losses, and improve disaster preparedness.
- **Sustainable agriculture**
Climate-smart farming techniques—including drought-resistant crops, efficient irrigation, soil conservation, and crop diversification—enhance food security and increase resilience to climate stressors.
- **Water resource management**
Integrated water management strategies, such as rainwater harvesting, watershed protection, and improved storage systems, help maintain water availability amid droughts, changing rainfall patterns, and growing demand.

IV. Activities and Assessment

Group Activity: Propose a School- or Community-Based Climate Action Initiative

Overview

In this activity, your group will design a **school- or community-based climate action initiative** that applies environmental science concepts and Information Systems (IS) solutions. Your proposal should address a real environmental problem and present actionable, feasible, and technology-supported strategies.

Instructions:

1. Group Formation

- Form groups of 6 members.
- Assign roles for each ensure equal contribution.

2. Activity Task

Each group must propose a climate action initiative that can be implemented at the school or in the local community.

Your proposal should:

- a. Identify a specific environmental issue (e.g., waste management, energy consumption, water conservation, carbon footprint reduction).

- b. Explain the relevance of the issue to the school or community.
 - c. Propose a climate action initiative that addresses the problem.
 - o Examples: campus solar charging stations, digital waste audit system, campus-wide zero-waste program, mangrove reforestation with GIS monitoring, climate awareness IS campaign, water consumption tracker app.
 - d. Integrate Information Systems concepts in your solution.
 - o Use of ICT tools, data management, automation, GIS, apps, dashboards, or online campaigns.
 - e. Provide a simple implementation plan including timeline, needed resources, and stakeholders.
 - f. Discuss expected environmental and social impacts of the proposed initiative.
 - g. Prepare A written proposal (2–3 pages) Format Written Proposal should include the following sections:
 - 1. Title of the initiative
 - 2. Introduction and background
 - 3. Identified environmental problem
 - 4. Proposed climate action initiative
 - 5. Information Systems integration
 - 6. Implementation plan
 - 7. Expected outcomes and impact
 - 8. References (APA format)
 - h. A 5–7 minute group presentation.
 - o Highlight the issue, proposed solution, IS integration, and impacts.
 - o Use digital slides (e.g., PowerPoint, Canva, Google Slides).
3. Submission
- Written proposal: Submitted via Google Classroom
 - Presentation: To be delivered in class during assigned schedule.

Rubrics for Evaluation

A. Written Proposal (50 points)

Criteria	Description	Points
Problem Identification	Clear explanation of the environmental issue and its relevance to school/community	10
Quality of Proposed Initiative	Feasible, innovative, and aligned with climate action goals	15
Integration of IS Concepts	Effective and appropriate use of ICT tools or IS frameworks	10
Implementation Plan	Clear steps, timeline, resources, and stakeholders identified	10
Organization & Writing Quality	Clear structure, grammar, coherence, proper APA citations	5
Total		50

B. Presentation (30 points)

Criteria	Description	Points
Content Clarity	Key points are well explained, concise, and informative	10
Presentation Design	Visual appeal, organization, and use of multimedia	5
Team Collaboration	Equal participation and smooth delivery	5
Understanding of IS Integration	Clear articulation of how IS enhances the initiative	5
Engagement	Ability to capture audience interest; question handling	5
Total		30